

M.AI.ESTRO VR: system audit

Finalized 4 October 2026 · Graham Pellegrini · University of Malta

System Demo



Demo recording — to be added

This will show the full orbit and task capture pipeline end to end.

Pipeline Diagram

The pipeline PNG shows the flow from headset capture to stored object evidence, with separate task and orbit branches. Open the full-resolution image to inspect individual labels; the PDF gives the complete diagram its own landscape A3 page.

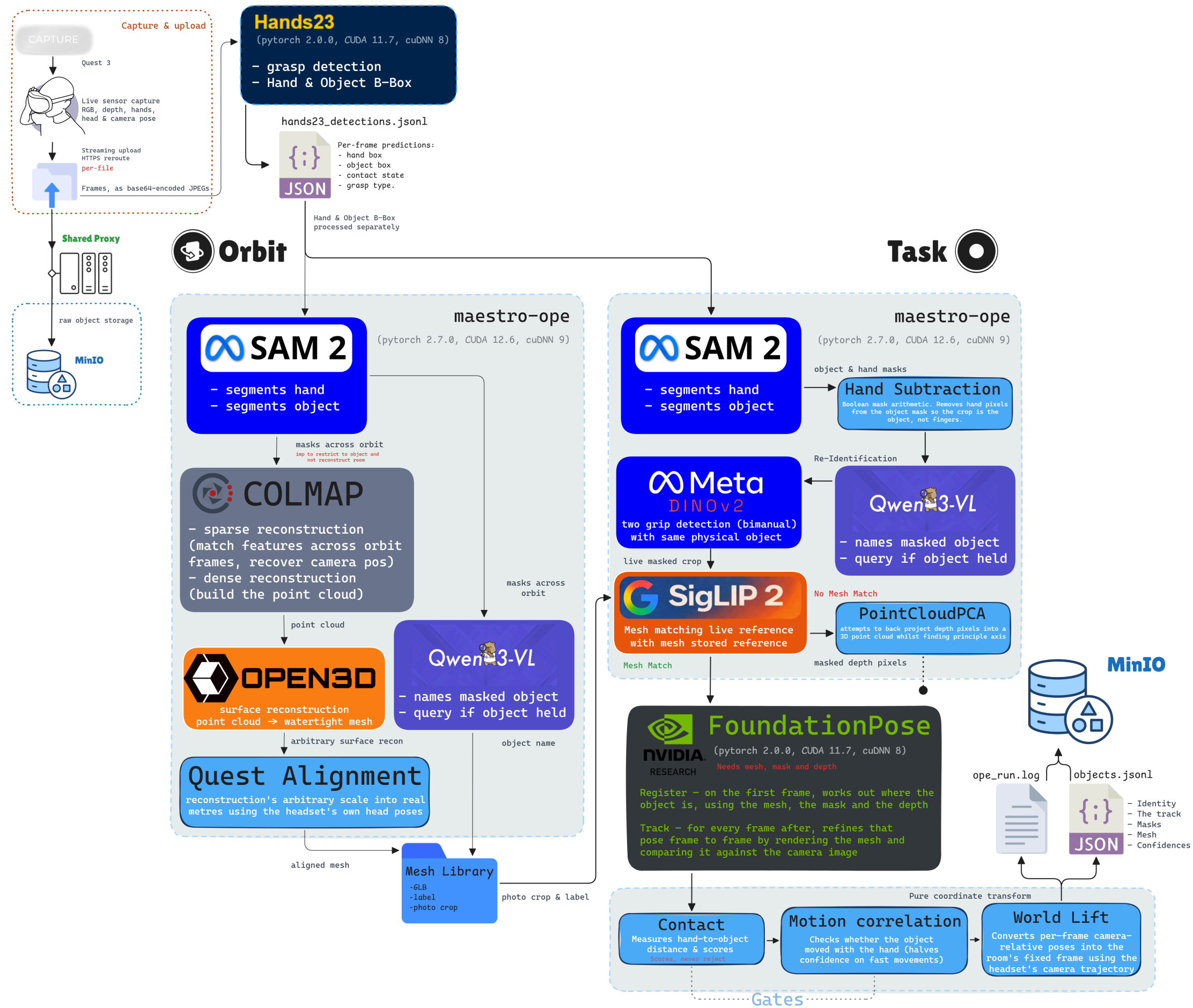


Figure 1. Current Maestros VR pipeline used for the audit discussion.

Pipeline Modules

Each node has a narrow responsibility. The audit question is whether the handoff between nodes is reliable enough for replay.

Quest capture

Records RGB, depth, headset pose and hand pose for task and orbit sessions.

Ingestion and MinIO

Accepts uploads, stores raw inputs and exposes session artifacts for worker processing.

OPE worker

Coordinates hand precompute, object evidence, masks, labelling, tracking and publication.

Hands23

Provides hand/object contact evidence and candidate boxes before object processing begins.

SAM2 masks

Turns candidate evidence into masks for orbit reconstruction and task-frame object tracking.

Qwen labels

Supplies an object label from the crop; useful, but still dependent on mask quality.

DINOv2 and SigLIP

Support visual identity checks. The current audit keeps DINOv2 and SigLIP rather than adopting DINOv3.

PointCloudPCA

Provides model-free task pose estimates when a trusted mesh is not available.

FoundationPose with mesh

Runs when a suitable mesh, mask and depth are available; this is the stronger route for mesh-backed replay.

Headset replay

Consumes stored objects, tracks, meshes and masks so captured objects can be inspected in VR.

System Status

The collapsed view is deliberately short for presentation. Open a card to see how the status was confirmed or what remains uncertain.

Confirmed and Working

Orbit route end to end

CONFIRMED

Hands23 to SAM2 to FoundationPose with a mesh is working for demo-grade orbit captures.



Episode naming endpoint

CONFIRMED

The ingestion naming path is in place for readable session identifiers.



Mesh visible in headset replay

CONFIRMED

Demo meshes are visible through headset replay for the selected sessions.



PointCloudPCA hand constraint guard

CONFIRMED

Impossible task poses during confirmed grip are rejected when they are more than 0.5m from hand landmarks.



Not Yet Confirmed

Task route with reliable pose

NOT CONFIRMED

PointCloudPCA without a mesh still needs stronger end-to-end validation.



Mesh matching

DISABLED

Automatic mesh matching is not accurate enough to trust.



DINOv3 consolidation

EVALUATED, NOT ADOPTED

The evaluation is complete, but production should keep DINOv2 and SigLIP.



Known Issues and Fixes

PointCloudPCA metre-scale jumps

PARTIALLY FIXED

Fast grip and contact-heavy motion can still expose task-pose instability.



ep067 bbox mismatch

FIXED

The stored bounding box did not match the served clock mesh dimensions.



Wireframe fallback without mesh

FIXED FOR DEMO SESSIONS

Replay without a trusted mesh can fall back to a less useful wireframe path.



Unity no-WiFi state

DESIGN FIX NEEDED

The headset UI shows connected and unstable states, but not a distinct fully offline state.



Technical Detail

Full technical audit and measurements

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